Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/04/2005
PI Name:	Boppart, Stephen A. Ph.D.		
Project Title:	A Nonlinear Optical Coherence Tomography System for Biomolecular Detection and Intervention		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	boppart@uiuc.edu	**Fax:**	FY 217-244-1995
PI Organization Type:	UNIVERSITY	**Phone:**	217-244-7479
Organization Name:	University of Illinois at Urbana-Champaign		
PI Address 1:	Beckman Institute		
PI Address 2:	405 N. Mathews Ave.		
PI Web Page:	http://nb.beckman.uiuc.edu/		
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/02/2002	**End Date:**	11/30/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
Contact Email:	pfung@mail.arc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dlott, Dana D. (UIUC) Gruebele, Martin H. (UIUC) Kitchell, Barbara E. (UIUC) Marks, Daniel L. (UIUC)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project involves the development of a new molecularly-specific optical imaging technique that we call Nonlinear Interferometric Vibrational Imaging (NIVI). NIVI combines and extends several imaging technologies, including Optical Coherence Tomography (OCT), OCT spectroscopic imaging, and Coherent Anti-Stokes Raman Scattering (CARS) spectroscopy, to produce a versatile system capable of noninvasively measuring micron-scale structure and molecular composition of tissues. These measurements can be made in real-time during treatment, potentially with a multiple frame-per-second acquisition rate. The high-peak-power pulses of the lasers used in this technique can also be used for selective tissue ablation, with subsequent monitoring . We will demonstrate the various high-resolution and spectroscopic modes of this instrument first on in vitro tissue models to verify operation, and then on an in vivo animal tumor model to determine its ability to detect specific biomolecules. NIVI is based on OCT, an imaging sensor that measures micron-scale tissue structure in vivo. It operates in a similar manner to ultrasound, by detecting optical reflections of laser light to range tissue density variations. Because of the comparatively small wavelength of light (about 1 micron) and the broad bandwidth laser sources available, micron-scale lateral and axial resolutions can be achieved, roughly 10-25 times better than ultrasound imaging. Because the laser pulses are of frequencies on the order of hundreds of terahertz, they can be shaped to resonantly stimulate the vibrational and rotational states of biomolecular groups. For example, the density of cell features such as peptide bonds, carbon-hydrogen bonds from cell membranes and lipids, carbonate and phosphate groups from bone hydroxyapatite, and potentially protein, RNA, and DNA biomolecules can be measured using Raman and Fourier-transform infrared spectroscopy. The combination of CARS and OCT will produce an instrument that is capable of micron-scale imaging of the molecular composition of tissues. With dynamic reshaping of the ultrafast probe pulses, we can address the resonance frequencies of various biomolecular species with the same instrument. The "coherent" nature of CARS allows axial discrimination of molecular density with OCT, as well as the ability to enhance the signal from a particular biomolecular group. This ability to select particular bond frequencies should enable much more precise density determination without background signals from other tissues. The coherent nature of CARS/OCT should also allow significantly greater sensitivity to signals generated from the tissues because of the coherent detection used in OCT. 1. Develop novel methods for achieving ultra-high-resolution OCT to image normal and cancerous tissue at the cellular level. 2. Develop spectroscopic OCT techniques to enhance contrast between tissue types. 3. Construct a unique and tunable laser system capable of performing NIVI. 4. Demonstrate NIVI for molecular-specific detection of pre-cursor moleculars associated with cancer. 5. Utilize high-energy laser pulses to ablate suspect cells in tissue. 6. Use ultra-high-resolution OCT, spectroscopic OCT, and NIVI to image ablation and monitor for re-growth of cancer. By developing and implementing novel technologies to generate, amplify, control, and detect ultrafast optical signals, this project will develop a method to recognize the signatures of specific biomolecular groups based on their morphological or spectroscopic features, or highly specific resonance frequencies. These optical methods will produce three-dimensional spatially-resolved structural and compositional data of tissues. Images are generated with low-power near-infrared light that harmlessly penetrates deep into highly-scattering tissue. Because NIVI is selective of a particular molecular species and employs coherent detection, extremely sensitive high signal-to-noise detection is possible. NIVI is non-contact, non-invasive, and readily applicable for in vivo imaging. Continued advances in photonic technology will decrease the size, weight, and power consumption of the NIVI components. The breadth of the available data will be applicable to many diseases, as well as disease-recognition algorithms. Finally, it will be possible to deliver focused high-peak-power laser pulses for selective ablation of diseased cells, and monitor subsequent results, providing a novel image-guided approach for detecting and treating cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progress on this project significant as we finalize the contract period. We have developed novel methods for interferometrically detecting and optically ranging the coherent CARS signal, permitting molecular-specific optical imaging. Additionally, we have developed new spectroscopic OCT algorithms that permit the extraction of depth-resolved absorption and scattering profiles from OCT data/images. These algorithms will fasciliate the detection of endogenous molecular changes in tissue, or the localization of exogenously contrast agents.
Bibliography Type:	Description: (Last Updated: 03/30/2006)
Articles in Peer-reviewed Journals	Marks, D. L.; Carney, P. S.; Boppart, S. A. "Adaptive spectral apodization for sidelobe reduction in optical coherence tomography images" J. Biomed. Opt. 2004 Dec;9:1281-1287 , Dec-2004
Articles in Peer-reviewed Journals	Mark, D. L.; Boppart, S. A. "Nonlinear interferometric vibrational imaging." Phys. Rev. Lett. 2004 Jan;92:123905. , Jan-2004
Articles in Peer-reviewed Journals	Vinegoni C.; Bredfeldt, J. S.; Marks, D. L.; Boppart, S.A. "Nonlinear optical contrast enchancement for optical coherence tomography" Optics Express 2004;12:331-341. , Jan-2004
Articles in Peer-reviewed Journals	Mark, D. L,; Vinegoni, C.; Bredfeldt, J. S.; Boppart, S. A. "Interferometric differentiation between resonant coherent anti-Stokes Raman scattering and nonresonant four-wave-mixing processes" Applied Phys. Lett. 2004 Dec;85:5787-5789 , Dec-2004
Articles in Peer-reviewed Journals	Xu, C.; Ye, J.; Marks, D. L,; Boppart, S. A. "Near-infrared dyes as contrast-enhancing agents for spectroscopic optical coherence tomography" Optics Letters 2004 Jul;29:1647-1649. , Jul-2004
Articles in Peer-reviewed Journals	Xu, C.; Boppart, S. A. "Separation of absorption and scattering profiles in spectroscopic optical coherence tomography using a least-square algorithm" Optics Express 2004 Sep;12:4790-4803. , Sep-2004

Articles in Peer-reviewed Journals	Xu, C.; Kamalabadi, F.; Boppart, S. A. "Comparative performance analysis of time-frequency distributions for spectroscopic optical coherence tomography" Applied Optics, In press, 2004. , Dec-2004
Articles in Peer-reviewed Journals	Bredfeldt, J. S.; Marks, D. L.; Vinegoni, C.; Boppart, S. A. "Molecularly sensitive optical coherence tomography" Optics Letters, In press 2004. , Dec-2004
Books/Book Chapters	Boppart, S. A. "Optical Coherence Tomography" in "Encyclopedia of Modern Optics" Ed. Guenther, R. D.; Steel, D. G.; Bayvel, L. Elsevier, 193-205 , Oct-2004